Vasanth Pidaparthy

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RESEARCH INTERESTS

My research lies in geometric analysis with a focus on fully non-linear elliptic PDEs. I work on problems at the interface of potential theory, a priori estimates, and differential and symplectic geometry.

EDUCATION

University of Maryland, College Park, MD
PhD in Mathematics | AUG 2021 – MAY 2027 (expected)
Advisor: Prof. Yanir A Rubinstein | Thesis: Weak and strong solutions to non-linear elliptic PDE

Stanford University, CA
Visiting Scholar | OCT 2025 – JUNE 2026

Indian Institute of Science, Bengaluru
– Master of Science in Mathematics | AUG 2020 – MAY 2021| First class w/ distinction
– Bachelor of Science (Research) in Mathematics | AUG 2016 – MAY 2020 | First class w/ distinction

BOOKS

Subequations | Graduate textbook | Accepted in AMS Graduate Studies in Mathematics (subject to revisions)
V. Pidaparthy, Y.A. Rubinstein | 301 pages | 2026
User's guide to the special Lagrangian equation | Graduate textbook (in preparation) | Solicited by the European Mathematical Society (EMS) | V. Pidaparthy

PUBLICATIONS AND PREPRINTS

Regularity via subequations | November 2026 | arXiv:2611.14253 | V. Pidaparthy, Y.A. Rubinstein
Convexity and the degenerate special Lagrangian equation | July 2025 | arXiv:2507.17018
V. Pidaparthy, Y.A. Rubinstein
Moduli of special Lagrangians with boundary, II: Lagrangian Flux and Affine Structures | Mar 2025 | arXiv:2503.23532 | V. Pidaparthy
Moduli of special Lagrangians with boundary, I: Unobstructed Deformations | to appear in Indiana University Mathematics Journal |arXiv:2503.23525 | V. Pidaparthy
The Hausdorff distance and metrics on toric singularity types | Bulletin des Sciences Mathématiques | Sep 2025 |A. Aitokhuehi, B. Braiman, D. Cutler, T. Darvas, R. Deaton, P. Gupta, J. Horsley, V. Pidaparthy, J. Tang

TALKS

(upcoming) **40th Geometry Festival** | UMD, College Park, Maryland, USA | APR 2–4, 2027
(upcoming) **Johns Hopkins Geometric Analysis Seminar** | Baltimore, Maryland | DEC 10, 2026
Geometry & Topology Seminar | IISc, Bengaluru, India | SEP 29, 2025

RESEARCH SUMMARY

Subequations and Regularity: Optimal regularity up to the boundary for solutions to the geometric PDE defined by a subequation, with applications to Monge–Ampère and special Lagrangian equations.

Special Lagrangians with boundary: Deformations of special Lagrangians with Lagrangian boundary conditions are unobstructed, and the corresponding moduli space admits a pair of dual special affine structures constructed using Lagrangian flux functionals.

DSL: Regularity up to the boundary fails in all branches of the degenerate special Lagrangian (DSL) equation under Lagrangian boundary conditions. Weak solutions exist for time-convex boundary conditions in the top two branches, and are (i) convex in the top branch, and (ii) time-convex in the second branch.

Interacting LLM Agents: Studying collective behavior in social networks of LLM-based agents and developing theoretical and mathematical approaches to characterize their dynamics (ongoing).

Structure in Mathematics: Developing information theoretic tools to study the geometry of mathematical results, modeled as a dependency acyclic graph. Goal is to understand when a new result reduces the `entropy' (ongoing).

AWARDS AND HONORS

Patrick and Marguerite Sung Fellowship in Mathematics | UMD | AUG 2025 – DEC 2025
Hauptman Summer Fellowship | UMD | MAY 2024 and MAY 2022
CMNS Outstanding Teaching Assistant Award | UMD | MAY 2022
Dean's Fellowship | UMD | AUG 2021 – MAY 2023
Kishore Vaigyanik Protsahan Yojana (KVPY) Scholarship | India | AUG 2016 – MAY 2021

LEADERSHIP AND TEACHING

Graduate Teaching Assistant, Department of Mathematics, University of Maryland, College Park
AUG 2021 – present (selected semesters) | **Responsibilities:** Classroom instruction, designing and grading quizzes and exams.

Graduate Teaching Assistant, Research Experience for Undergraduates (REU)
– JUN 2024 — JUL 2024 | Convex and Complex Geometry | 6 undergrads | Publication (Bull. Sci. Math., 2025)
– JUN 2023 — JUL 2023 | Geometric Measure Theory | 4 undergrads

Mentor, Directed Reading Program (DRP) | FEB 2023 – MAY 2023 | Topic: The Bolzano–Weierstrass theorem

Instructor, Monsoon Math Camp | JUN 2021 | Online high school math camp in India | Probability, martingales, and gambler's ruin

Co-organizer, Student Complex Geometry Seminar, UMD | AUG 2023 – MAY 2024
Delivered a semester-long lecture series on Gromov–Hausdorff convergence of Ricci lower bounded manifolds.

Social Coordinator, Graduate Student Council of MATH, AMSC, and STAT | AUG 2023 – MAY 2024
Created and organized the JAMS student seminar to give grad students exposure to broad fields of mathematics

INTERNSHIP EXPERIENCE

Quantum Computing Research Intern, Error Corp. | JUN 2025 — AUG 2025 | 10 weeks (Full time)
– Modelled the problem of minimizing first order interaction/cross-talk errors in a 2-qubit quantum system as a differential geometry problem of identifying optimal control curves in the lie group SU(2).
– Implemented a numerical solver in Python (JAX) to optimize control pulses, achieving a significant reduction in error rates. Delivered a Python repository.